Project Report

Quantum State Tomography: Reconstruction of Unknown Quantum States using Quantum Measurements and Maximum Likelihood Estimation

Roohi Verma, 2nd Physics Maths Year Undergraduate

Project Supervisor: Prof. Arvind

Project Duration: 15th May 2025 - 13th August 2025

Abstract

In Quantum Computation and Information Theory there is an extensive use of Quantum Computers. Similar to Classical computers having bits, QCs have Qubits which store information regarding a Quantum State. Sometimes, the quantum state associated with the qubit is not known. We use statistical analysis such as the maximum likelihood estimation to assume parameters and to generate a function called the likelihood function which expresses the likelihood of generating the observed data given the parameters that we have assumed in the model. This likelihood is then maximized in order to get the values of parameters. The values of the parameters are estimated using python packages and a demonstration of the whole algorithm involves the use of python packages like Qiskit. Furthermore, a brief account on a more manual mathematical approach is given using the bloch vector representation. The advantage of Maximum Likelihood Estimation is given over this method.

13/8/25

Project Supervisor
(Prof Arvind)